

# PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

## Module 9 Exam Notes: Microwave Systems & Modern Trends

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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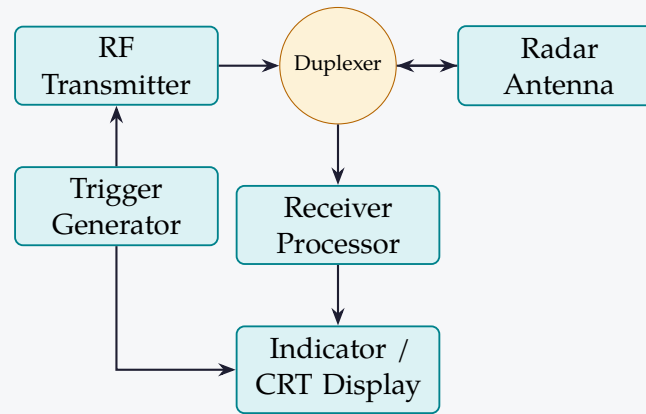
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## 1. Radar Systems

RADAR (Radio Detection And Ranging) is an active electromagnetic system that transmits microwave pulses and processes reflections to determine target range, velocity, and angle.

**Pulsed Radar Block Diagram**



### Radar Range Equation

The maximum radar detection range  $R_{\max}$  is given by:

$$R_{\max} = \left[ \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{\min}} \right]^{1/4} \quad (\text{m}) \quad (1)$$

Where:

$P_t$  = peak transmitter power (W)

$G$  = antenna gain

$\lambda$  = RF operating wavelength (m)

$\sigma$  = Radar Cross Section (RCS) of target ( $\text{m}^2$ )

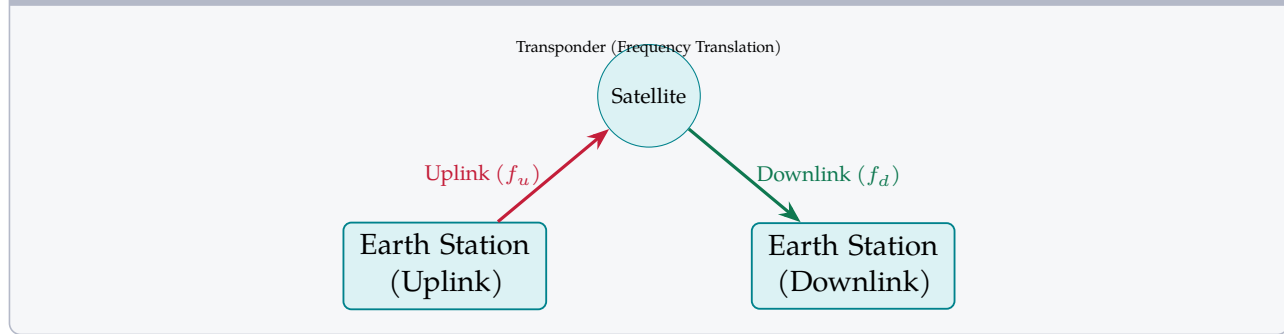
$P_{\min}$  = minimum detectable receiver signal (W)

### Radar Variations:

1. **CW (Continuous Wave) Radar:** Transmits continuously, measures velocity using Doppler shift ( $f_d = 2v/\lambda$ ).
2. **FMCW (Frequency Modulated CW) Radar:** Sweeps transmitter frequency, allowing both range and velocity to be measured.
3. **MTI (Moving Target Indicator) Radar:** Suppresses static clutter (ground, buildings) using delay-line cancelers.

## 2. Terrestrial and Satellite Communications

### Satellite Uplink/Downlink Path Schematic



**Link Budget ( $C/N$ ):** Carrier-to-Noise ratio at the receiver terminal:

$$\left(\frac{C}{N}\right)_{\text{dB}} = \text{EIRP}_{\text{dBW}} + \left(\frac{G}{T}\right)_{\text{dB/K}} - \text{FSL}_{\text{dB}} - k_{\text{dBW}} - B_{\text{dB-Hz}} \quad (2)$$

where EIRP is Equivalent Isotropic Radiated Power,  $G/T$  is receiver figure of merit,  $\text{FSL} = (4\pi d/\lambda)^2$  is Free Space Path Loss, and  $k$  is Boltzmann's constant.

### 3. Radio Aids to Navigation

- **GPS (Global Positioning System):** Employs 24+ satellites. Receivers measure transit times of signals from 4+ satellites, using **triangulation** to resolve latitude, longitude, altitude, and clock bias.
- **ILS (Instrument Landing System):** Provides guidance for aircraft landing: **Localizer** (lateral guidance, 108–112 MHz) and **Glide Slope** (vertical guidance, 329–335 MHz).
- **LORAN (Long Range Navigation):** Hyperbolic radio navigation system using time-difference arrivals of pulses from ground stations.
- **RFID (Radio Frequency Identification):** Uses electromagnetic coupling in microwave/RF bands for asset tracking.

### 4. Modern Trends: Biological Effects, MMICs, and RFMEMS

- **Biological Effects & SAR:** Microwaves cause dielectric heating of tissues. The **SAR (Specific Absorption Rate)** measures RF energy absorption by body tissues:

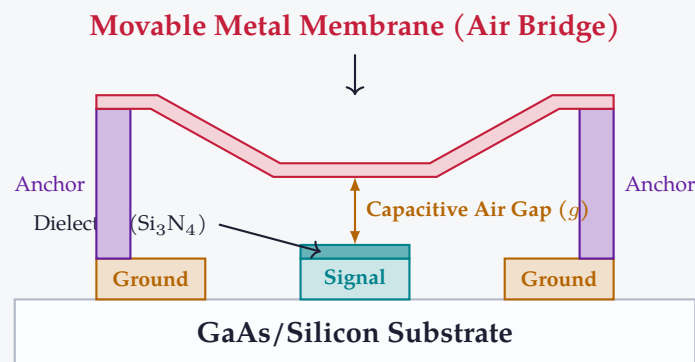
$$\text{SAR} = \frac{\sigma |E|^2}{\rho} \quad (\text{W/kg}) \quad (3)$$

where  $\sigma$  is tissue conductivity,  $E$  is electric field, and  $\rho$  is mass density. Regulatory limits are typically 1.6 W/kg over 1 g of tissue.

- **MMICs:** Monolithic Microwave Integrated Circuits integrate passive components, transmission lines, and active transistors onto a single semiconductor substrate (GaAs, GaN, SiGe).

- **RFMEMS:** Micro-Electro-Mechanical Systems designed for RF/microwave signals. They offer high isolation, low insertion loss, and low power consumption.
- **Microwave Imaging:** A non-invasive imaging technique that maps the spatial distribution of dielectric properties (permittivity and conductivity) inside a body.
  - **Active Imaging (Radar-based):** Transmits low-power microwave pulses and records the backscattered signals to reconstruct a 3D profile of dielectric contrasts.
  - **Passive Imaging (Radiometry):** Detects natural thermal microwave radiation emitted by the body using ultra-sensitive radiometers.
  - **Applications:** Non-destructive testing, airport security scanners (millimeter-wave body scanning), and medical diagnostics (such as breast cancer screening, which leverages the 5:1 permittivity contrast between high-water-content tumors and normal fat tissue).

#### RFMEMS Capacitive Switch Model



*Cross-sectional view. Electrostatic pull-down voltage collapses the bridge onto the dielectric, switching state from OFF to ON.*

## Quick Revision — Module IX

Microwave Systems & Modern Trends | PE-EC701A

### Key Formulas at a Glance

|                                 |                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------|
| <b>Radar Range Equation</b>     | $R_{\max} = \left[ \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{\min}} \right]^{1/4}$ |
| <b>Doppler Frequency Shift</b>  | $f_d = \frac{2v}{\lambda} \cos \theta$                                               |
| <b>Free Space Path Loss</b>     | $\text{FSPL} = \left( \frac{4\pi d}{\lambda} \right)^2$                              |
| <b>Specific Absorption Rate</b> | $\text{SAR} = \frac{\sigma  E ^2}{\rho}$                                             |
| <b>Shielding Effectiveness</b>  | $\text{SE}_{\text{dB}} \approx A_{\text{dB}} + R_{\text{dB}} + M_{\text{dB}}$        |

### One-Line Definitions

- **Duplexer:** A fast waveguide switch permitting a single antenna to be shared by radar transmitter and receiver.
- **Doppler Effect:** The change in frequency of a received wave due to relative motion between source and observer.
- **SAR:** Specific Absorption Rate, measuring the rate of electromagnetic energy absorption per unit mass of human tissue.
- **MMIC:** Monolithic Microwave Integrated Circuit, fabricated entirely on a single semiconductor wafer.
- **RFMEMS:** Micro-Electro-Mechanical Systems utilized for high-performance microwave switching and tuning.
- **Triangulation:** Geometric position mapping utilizing measured transit distances from multiple satellites.
- **Clutter:** Unwanted reflections from ground objects, weather, or buildings that interfere with target radar signals.
- **Microwave Imaging:** A non-invasive imaging method mapping dielectric property contrasts within a target.

### Frequently Asked / Viva Questions

#### Q1. Why is a duplexer critical in pulsed radar systems?

It isolates the sensitive receiver from the high-power transmitter during the pulse transmission, and routes the weak return echo to the receiver during the rest interval.

#### Q2. How does Doppler shift change with target movement direction?

If the target moves toward the radar, the received frequency increases  $(+f_d)$ . If it moves away, the received frequency decreases  $(-f_d)$ .

**Frequently Asked / Viva Questions (contd.)****Q3. Explain the term "Specific Absorption Rate" (SAR).**

It is the power absorbed per unit mass of human tissue under RF exposure, measured in W/kg, used to define safety standards for mobile devices.

**Q4. What are the advantages of MMICs over hybrid Microwave Integrated Circuits (HMICs)?**

Small size, low weight, high reliability, ease of mass production, and elimination of wire bond parasitics.

**Q5. Describe the function of localizer and glide path transmitters in ILS.**

The localizer provides horizontal guidance relative to the runway centerline. The glide path provides vertical descent guidance to the runway touchdown point.

**Q6. Why does Free Space Path Loss (FSPL) increase with frequency?**

Because effective antenna aperture decreases as frequency increases ( $A_e \propto 1/f^2$ ), reducing the power intercepted by a matched receiving antenna.

**Q7. What are blind speeds in MTI radar and how are they avoided?**

Blind speeds are target velocities where the Doppler shift matches the radar pulse repetition frequency, making targets appear static. They are avoided by using staggered PRFs.

**Q8. What materials are primarily used to fabricate MMICs and why?**

Gallium Arsenide (GaAs) and Gallium Nitride (GaN) due to high electron mobility, which is necessary for gigahertz operation.

**Q9. How does a capacitive RFMEMS switch operate?**

An electrostatic voltage pulls a suspended metal bridge down onto a dielectric-covered CPW line, dramatically increasing coupling capacitance to short the RF signal.

**Q10. Explain how GPS resolves a 3D position using four satellites.**

Three satellites locate the receiver at the intersection of three spheres. The fourth satellite resolves the clock synchronization error between the receiver and GPS time.

**Q11. State the thermal effects of microwaves on the human body.**

Dielectric heating of water molecules in body tissues, which can lead to localized temperature increases and tissue damage (especially in the eyes and testes).

**Q12. What is shielding effectiveness (SE) and what are its main mechanisms?**

It measures a barrier's ability to attenuate electromagnetic fields, governed by absorption loss, reflection loss, and multiple-reflection corrections.

**Q13. Differentiate between active and passive microwave imaging.**

Active microwave imaging transmits low-power microwave pulses and measures backscattered signals to reconstruct a dielectric profile. Passive microwave

**Frequently Asked / Viva Questions (contd.)**

imaging uses highly sensitive radiometers to detect the natural thermal electromagnetic radiation emitted by the body.

**Q14. Why is microwave imaging promising for breast cancer detection?**

Because malignant tumors have a high water content and thus show a significant dielectric contrast (permittivity and conductivity are about 5 times higher) compared to normal healthy breast fat tissue. This allows clear imaging without ionizing X-ray radiation.

**Comparison of MMICs and Hybrid MICs**

| Parameter             | Monolithic Microwave ICs (MMICs)                               | Hybrid Microwave ICs (HMICs)                                    |
|-----------------------|----------------------------------------------------------------|-----------------------------------------------------------------|
| Substrate             | Single semiconductor wafer (GaAs, GaN).                        | Alumina, quartz, or Teflon board.                               |
| Component Integration | Transistors and matching lines are grown in-situ on the wafer. | Discrete chips are soldered or wire-bonded to substrate tracks. |
| Reliability           | Extremely high (no wire bonds to fail).                        | Moderate (wire bonds are vulnerable).                           |
| Cost (Low Volume)     | Very High (complex mask tooling).                              | Moderate (easy prototyping).                                    |
| Cost (High Volume)    | Very Low (mass batch production).                              | High (requires manual/auto assembly).                           |